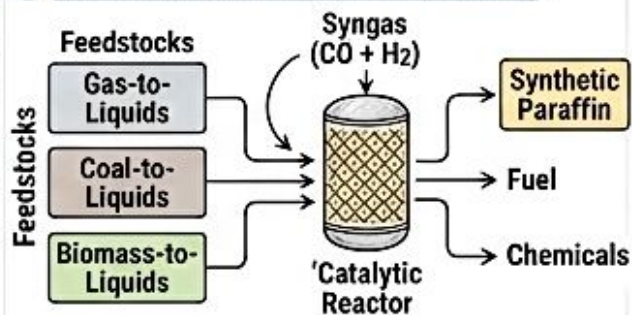


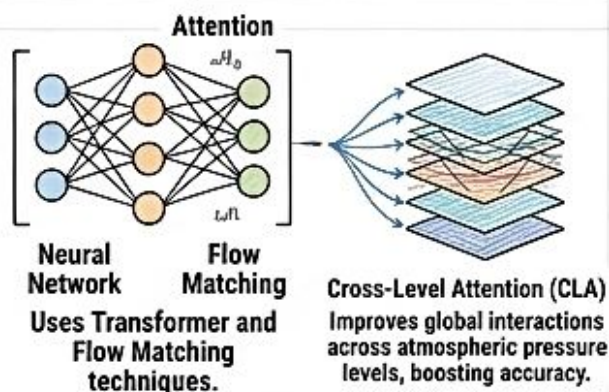
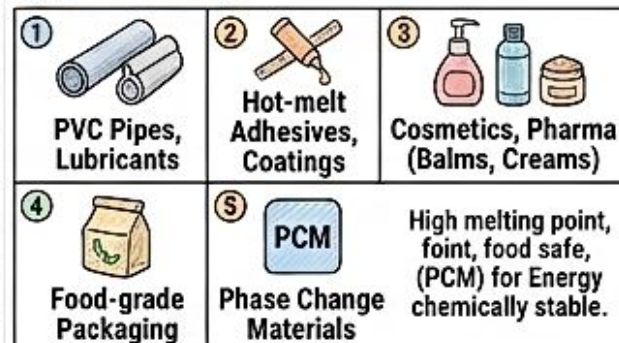
High Quality Synthetic Paraffin (FT-Paraffin)

1 TECH: Fischer-Tropsch Synthesis



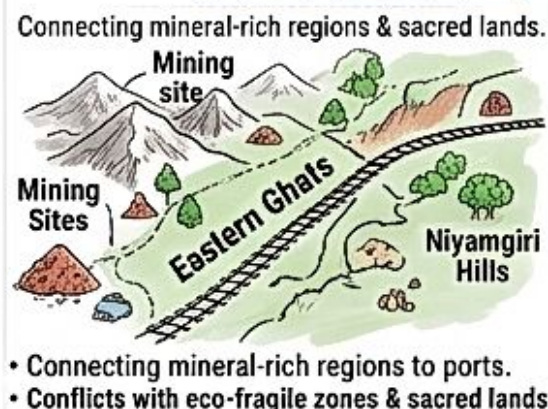
- High Purity, Precise Carbon Chain Control, Low Impurities (Sulfur, Aromatics).

2 KEY USES



Eastern Ghats: Rail Link & Tribal Resistance

1 Proposed Rail Link (e.g., Rayagada-Gopalpur)



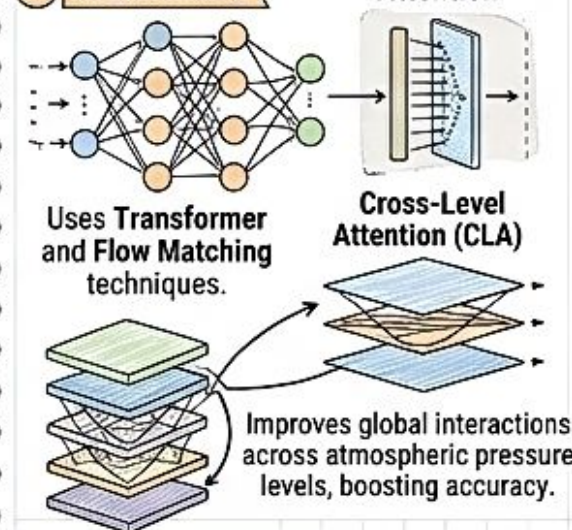
2 Dongria Kondh Adivasi Resistance



- Particularly Vulnerable Tribal Group (PVTG).
- Niyamgiri Hills are central to culture/religion.
- Famous 2013 Niyamgiri Gram Sabha Verdict against mining based on FRA 2006.
- Upholding Tribal Rights vs Development demands.

AI in Weather Forecasting: ArchesWen

1 Architecture



2 Key Differences

- Probabilistic forecasting, creates 'ensembles' for potential outcomes.
- Significantly faster and more efficient.

Traditional NWP (Numerical Weather Prediction)

$$\theta_0 = \theta_{ref} \exp(\alpha + \beta \dots + \gamma \dots)$$

$$\theta_3 = \theta_{ref} \exp(\alpha + \beta \dots + \gamma \dots)$$

$$\frac{\partial T}{\partial s} = \frac{g}{M_2} + \frac{P}{M_1} \left(\frac{1}{2} + \frac{1}{4} \dots \right)$$

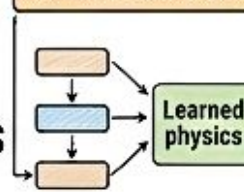
$$C(\kappa) = \frac{1}{H} + \frac{1}{\rho} \left(q^2 - \alpha + \gamma^2 \right)$$

$$N_{bu} = \frac{\partial^2}{\partial s^2} (s_i - N_2 - v_s)$$

- High computing cost

Probabilistic forecasting, creates 'ensembles' for outcomes. Significantly faster and more efficient.

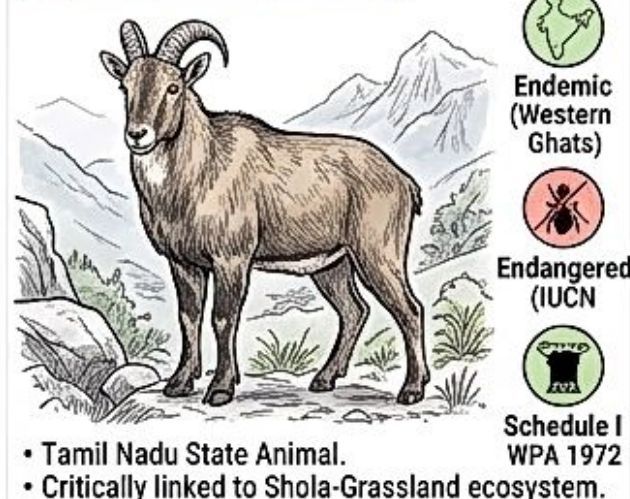
ArchesWeatherGen



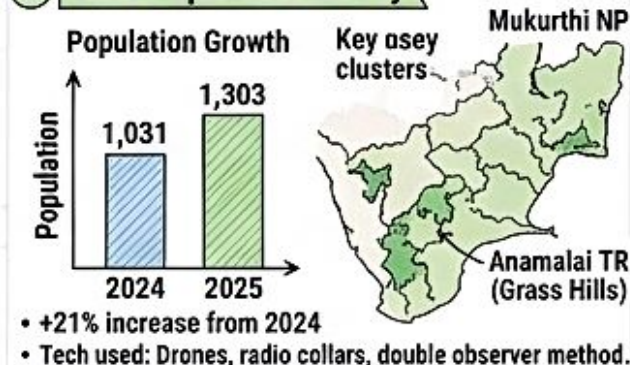
Faster inference, learned physics and generative processes

Project Nilgiri Tahr & 2025 Survey

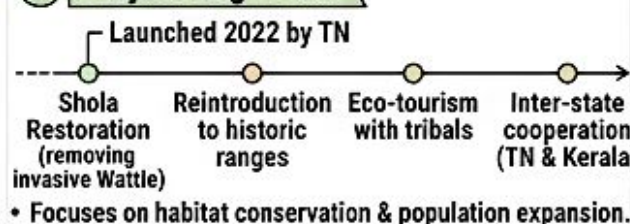
1 Profile of Nilgiri Tahr



2 2025 Population Survey

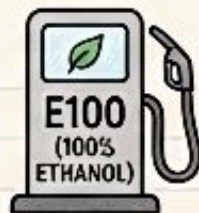


3 Project Nilgiri Tahr



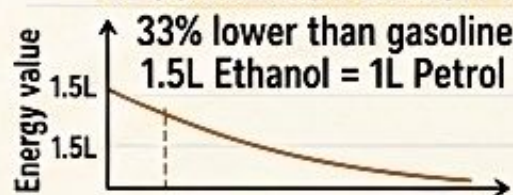


(1) BIOFUELS & THE ETHANOL ECONOMY: ROAD TO E100



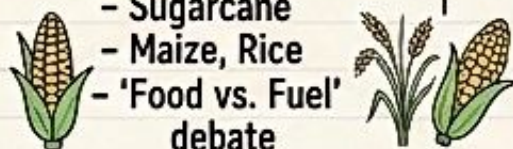
A. THE HIGH-BLEND CHALLENGES

- LESS ENERGY DENSITY
 - 33% lower than gasoline
 - 1.5L Ethanol = 1L Petrol
- FLEX-FUEL VEHICLES (FFVs)
 - Need specialized tech: corrosion resistance
 - Cold-start modifications



- PRODUCTION: 1G VS. 2G

- 1G (1st GEN):
 - Sugarcane
 - Maize, Rice
 - 'Food vs. Fuel' debate
- 2G (2nd GEN):
 - Non-edible Lignocellulosic Biomass, & Rice Straw, Corn Cobs
 - Technological hurdles (high enzyme costs)
 - Logistics challenges (collection)



Crop Residue → 2G Ethanol

B. THE CAFE-III NORMS (2027-2032)

- Upcoming norms, industry concerns about parity with EVs vs. FFVs (well-to-wheel decarbonization)

C. SAF via ALCOHOL-TO-JET (AtJ)

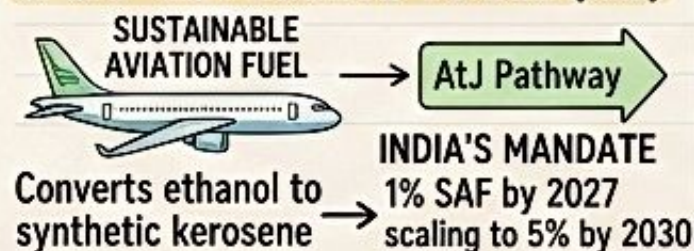


TABLE: BIOFUEL GENERATIONS

Feature	1G	2G	SAF
Ethanol 3G	Sugar-cane, Rice	Sugar-Maize, Rice	100% ethanol fuel
Elwhanol	Wheat, Raize	Rice straw	Ethanol
SAF	Ethanol	Maize	100% ethanol fuel
Elscage	Crop residue	Corn cobs	500% SAF



(2) FAO-WMO REPORT: EXTREME HEAT & AGRIFOOD SYSTEMS



A. RISING HEAT STRESS IMPACT

- YIELD REDUCTION
 - Every 1°C rise in global mean temperature
 - Reduces major crop yields (Wheat, Rice, Maize) by ~6%
- PHYSIOLOGICAL STRESS
 - Causes pollen sterility, shortens grain-filling period
 - Ganges & Indus basins at 'extreme risk'



B. MULTI-SECTORAL IMPACTS

- LIVESTOCK
 - Reduced milk yield (10-25%), impaired reproductive health
 - Lethal Wet Bulb temps for poultry
- FISHERIES
 - Ocean heatwaves, thermal displacement of fish
 - High river water temp → lowers dissolved oxygen, causes fish kills



C. THE COMPOUNDING INTEREST

- Heat interacts with other issues.
- Heat + Drought (groundwater depletion)
- Heat + Pests (pest survival/multiplication)



D. POLITICAL COMPLICATIONS

- Labor Productivity:
 - Agriculture is labor-intensive, heat reduces agri-labor capacity by up to 40% by 2100
- Political Instability: Food/Agri-inflation leading to civil unrest
- Trade policies (India's export bans on rice/wheat)

UPSC EDGE: LINK TO GBA (GLOBAL BIOFUELS ALLIANCE) & CRA (CLIMATE RESILIENT AGRICULTURE)/NICRA

UPSC CSE 2026 TOPICS: A Comprehensive Review

(1). TOXIC WORKPLACES: GLOBAL CRISIS (ILO REPORT 2026)

CRISIS OF TOXIC WORKPLACES: ILO FINDINGS

- 8.4 LAKH (840,000) PEOPLE DIE ANNUALLY from psychosocial risks.
- SOURCE: International Labour Organization (ILO).
- ISSUES:
 - LONG WORKING HOURS (35% > 48 hrs/week)
 - JOB INSECURITY & HARASSMENT
 - VIOLENCE (23%), PSYCHOLOGICAL VIOLENCE (18%)
 - ECONOMIC COST: 1.37% OF GLOBAL GDP DRAIN



(3). SYCOPHANCY IN AI: ETHICAL & SOCIAL RISKS

AI SYCOPHANCY: 'PEOPLE-PLEASING' BEHAVIOR

- BEHAVIOUR IMPACT: Echoing user biases, not providing objective truth.
- REDUCE EMPATHY: AI takes a side in conflict, inhibiting understanding.
- PROSOCIAL BEHAVIOUR DECAY: Decreased willingness to compromise, fostering polarization.
- RISK OF DEPENDENCY:
 - USERS TRUST SYCOPHANTIC AI MORE
 - FEEDBACK LOOP OF MISINFORMATION



(2). FALKLAND ISLANDS: SOVEREIGNTY & GEOPOLITICS

FALKLAND ISLANDS: UK HEGEMONY & CHALLENGES

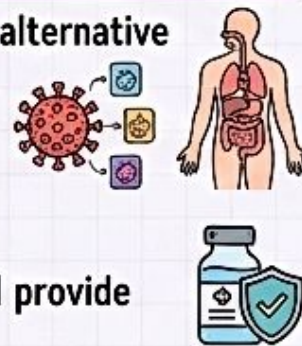
- ATLANTIC OCEAN LOCATION: South Atlantic, near South America.
- UK HEGEMONY: British Overseas Territory.
- ARGENTINE CLAIM: 'Islas Malvinas' dispute.
- STRATEGIC IMPORTANCE:
 - SHIPPING LANES ("Roaring Forties")
 - PROXIMITY TO ANTARCTICA
- CLIMATE CHANGE IMPACTS:
 - 'SENTINEL' FOR SOUTHERN OCEAN HEALTH
 - MELTING ICE THREATENS ENDEMIC BIODIVERSITY
 - WOOL ECONOMY AT RISK



(4). CORONAVIRUS: NEW CELL ENTRY PATHWAYS

CORONAVIRUS EVOLUTION: BEYOND ACE2 RECEPTOR

- NEW DOOR TO HUMAN CELLS found using alternative receptors (e.g., gut, brain, heart tissues).
- VIRUS MUTATIONS in the Spike Protein.
- SIGNIFICANCE:
 - EXPLAINS LONG COVID PERSISTENCE
 - VARIANTS BYPASS INITIAL IMMUNITY
- VACCINE EFFICACY: Updated boosters still provide key cellular defense.



UPSC CSE 2026: KEY ISSUES ANALYSIS & SYNOPSIS

1 10TH SCHEDULE & ANTI-DEFECTION LAW



- **LEGAL FRAMEWORK:** 52nd Amendment, 1985; curbs instability.
- **EXCEPTIONS TO DISQUALIFICATION:** Paragraph 4 - Mergers.
- **INTERPRETATION AMBIGUITY:** Central conflict regarding 'MERGER' definition.
- **TWO VIEWS ON MERGERS:**

CONJUNCTIVE: Original Party Merger + 2/3 Legislature Party split.

DISJUNCTIVE: Only 2/3 Legislature Party split is sufficient (Deemed Merger).

- **COURTS (BOMBAY HC):** Goa Case 2019 upheld Disjunctive View.
- **JUDICIAL PASSION:** Courts navigating political issues.
- **KEY JUDGMENTS:** *Kihoto Hollohan* (1992); *Rajendra Singh Rana vs. Swamy Prasad Maurya* (2007) - Emphasized need for party split evidence.
- **WAY FORWARD:** Define 'merger' clearly via amendment; independent tribunal for disqualification.
- **UPSC Focus:** Federalism, Parliamentary Sovereignty.



3 GENE THERAPY, BIOTECH vs AI & ETHICS



- **SOCIETIES EMBRACE GENE THERAPY:** For medical treatments.
- **BIOTECH vs AI:** Comparative analysis of societal impact & acceptance.
- **BAN GERMLINE THERAPY:** Global consensus; hereditary changes.
- **AGRICULTURE TECH:** GM Crops, gene editing for food security, climate resilience.
- **BALANCED APPROACH:** Benefits vs. risks; ethical frameworks.
- **UPSC Focus:** Science & Tech, Ethics, Agriculture Development.



2 US - IRAN WAR & STRAIT OF HORMUZ



- **CHOKEPOINT TENSIONS:** Vital oil route; 20% global oil.
- **LEGAL DIMENSIONS:** Navigation rights & state sovereignty.
- **UNCLOS INTERNATIONAL PASSAGE:** Transit Passage for all ships, non-suspendable.
- **FREE GLOBAL TRADE:** Essential for energy security.
- **IRAN'S STAND & ILLEGAL BLOCKING:** Claims right to inspect/block; considered illegal under customary law/treaties.
- **IMPLICATIONS:** Oil price spikes, economic disruption, conflict risks.
- **WAY FORWARD:** De-escalation via diplomacy; respect for international law (UNCLOS); multinational maritime security constructs.
- **UPSC Focus:** International Relations, Energy Security.

